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SIMATS
ENGINEERING

CO4-AT2 – Coding Golf Contest

QUESTIONS

01
Quick Sort
25 marks

01 Quick Sort

SIMATS Online Programming Lab

DS PROGRAMMING

H Harshavardhan
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Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort

02

Bubbled Sorting

25 marks

Q3.

Insertion Sort

25 marks

Q4

Merger Sort

25 marks

Your Code

```
#include <stdio.h>

2 void swap(int* a, int* b) {
    int t;

    *a = *b;
    *b = *a;
    *a = t;
}

int partition(int arr[], int low, int high) {
    int pivot = arr[low];

    int i = low + 1;
    int j = high;

    while (i <= j) {
        while (i <= j && arr[i] <= pivot) i++;
        while (j > i && arr[j] > pivot) j--;
        if (i < j) swap(&arr[i], &arr[j]);
    }
    swap(&arr[low], &arr[j]);
    return j;
}

int main() {
    int arr[] = {10, 20, 30, 40, 50, 60, 70, 80, 90, 100};
    int low = 0, high = 9;
    int partition_index = partition(arr, low, high);
    printf("Partition index: %d\n", partition_index);
    return 0;
}
```

4/4 answered

```
for (int) low; j<=high- 1; j++){

12
    if (arr[] pivot) {
13
14
        swap (saxx[1], saxx[1]);
15
        2
16
    }

18

19 1
    sump (l1), Garr(high));
    return (i + 1);

20 void quicksort (int arr[], int low, int high) (
    if (low <high){
        int pi
            partition(arr, low, high);
        quicksort (arr, low,pi - 1);
21
22
23
24
25
26)
27 int main() {
    quicksort (arr, p1 +1, high);
```

Input

stdin input

Output

18 23:32 55 61

← Back

25 marks

All Bookmarks

02

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02

Your Code

Bubbled Sorting

SIMATS Online Programming Lab

DS PROGRAMMING

25 marks

Q3

Insertion Sort

25 marks

Q4

Mejer Sort

25 marks

4/4 answered

```
#include <stdio.h>

2 void swap(int* a, int* b) {
    int t = *a;
    *a = *b;
    *b = t;
}

int partition (int arr[], int low, int high) {
    int pivot = arr[high];

    for (int i = low; i <= high-1; i++) {
        if (arr[i] < pivot) {
            swap (&arr[i], &arr[high]);
        }
    }

    swap (&arr[high], &arr[low]);
    return (low);
}

20 void quicksort (int arr[], int low, int high) {
    if (low < high) {
        int pi = partition (arr, low, high);
        quicksort (arr, low, pi - 1);
        quicksort (arr, pi + 1, high);
    }
}

27 int main() {
    int arr[] = {150, 23, 9, 18, 61, 321};
    int n = sizeof(arr)/sizeof(int);

    quickSort (arr, 0, n-1);

    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    printf("\n");
    return 0;
}
```

Input

stdin input

Output

18 23:02 50/61

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192511171RD

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SIMATS
ENGINEERING

QUESTIONS

Q1
Quick Sort
25 marks

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DS PROGRAMMING

H Harshavardhan
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Q2 Bubble Sort

Write a program that implements the bubble sort.

25 marks

Q2
Bubble Sort
25 marks

Q3
Insertion Sort
25 marks

Q4
Merge Sort
25 marks

Your Code

```
#include <stdio.h>

3 void bubbleSort (arr, n) {
    for (int i=0; i<n-1; i++) {
        int swapped = 0;
        for (int j=0; j<n-i-1; j++) {
            if (arr[j]>arr[j+1]) {
                int temp = arr[j];
```

4/4 answered

```
        arr. -]
        accll= temp;
        swapped=1;
11
12
13
14         if (swapped) break;
15
16
161
17 int main() {
18     int n;
19
20     int a[n];
21
22
23     J
24     scanf ("%d", &n);
25
26     for (int i = 0; i < n; i++) {
27         scanf ("%d", &arr[i]);
28
29         bubbleSort (arr, n);
30     for (int i=0; i < n; i++) {
31
32     }
33
34     printf ("%d", arr[1]);
35
36     return 0;
37 }
```

Input

5
22245

Output

12345

All Bookmarks

QUESTIONS

Q1
Quick Sort
25 marks

Q2
Bubble Sort
25 marks

Q3
Insertion Sort
25 marks

Q4
Merge Sort
25 marks

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Q3 Insertion Sort

DS PROGRAMMING

Write a program sort the elements 9, 5, 1, 4, 3 using Insertion Sort.

Your Code

```
#include <stdio.h>

2 void insertionSort(int arr[], int n) {
    int i, key, j;
    for (i = 1; i < n; i++) {
        key = arr[i];
        j = i - 1;
        while (j > 0 & arr[j] > key) {
            arr[j+1] = arr[j];
```

Input

stdin input

Output

13459

4/4 answered

```
11         arr[j+1] = key;
12     }
131
14 void printArray(int arr[], int n)
15 {
16     for (i = 0; i < n; i++) {
17         printf("%d ", arr[i]);
18     }
19     printf("\n");
20 int main() {
    int arr1[9] = {9, 5, 1, 4, 3, 17};
    int n = sizeof(arr1) / sizeof(arr1[0]);
22
23
24     return 0;
    InsertionSort(arr1, n);
    printArray(arr1, n);
}
```

Submitted

25 marks

All Bookmarks

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QUESTIONS

Q1
Quick Sort
25 marks

Q2
Bubble Sort
25 marks

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Q4 Merger Sort

DS PROGRAMMING

Apply Merge Sort to the array 38, 27, 43, 3, 9, 82, 10.

Q3
Insertion Sort
25 marks

Q4
Merger Sort
25 marks

```
Your Code
#include <stdio.h>
2 void merge(int arr[], int l, int m, int r){
    int i, j, k;
    int a[] = arr;

3
    int a2[] = L;
    int L[n/2], R[n/2];

    for (l = 0; l < n; l++)
        L[l] = arr[l + 1];

9
    for (l = 0; l < n; l++)
        R[l] = arr[l + 1];
```

4/4 answered

```
11         [i+1 + j]);
12         l = 0;
13
14         l = 0;
15
16         while (<1 663 m2) 1
17             if ((i)R(j)) {
18                 r[k]= [];
19
20                 1 else {
21                     [k]= [];
22                     3++8
23
24                     while (l < 0) {
25                         are [k]=L[];
26                         2417
27                         3417
28
29                     while (52) {
30                         are [x] = R[j];
```

Input

stdin input

Output

38:10 27 38 43 82

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25 marks

All Bookmarks

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```

    } else {
        OFF[k-1];
120
21
22
23
24
    while (l) {
25
        are [x] = {1};
26
        l++;
27
        3++7
28
    }
29
30
    while (<2) {
        acc[k] = R[J];
31
        3
32
33
    }
241
55 void mergeSort (int arr[], int l, int r)
(
36 if (l < r) (
38
39
40
41
        7
421
        int m = l + (l) / 2;
        merge Sort. (arr, l, m);
        merge Sort (arr, +1, );
        merge (arr, l, r, l);

49 int main() (
44 int arr[] = {38, 27, 43, 3, 9, 82, 10};
45
        int arr size = sizeof(arr)
                                sizeof(arr[0]);
46
        mergesort (arr, 0, arr size-1);
        for (int i = 0; i < arr_size; i++){
58
            printf("d", arr[i]);
49
        )
50
        printf("n");
        return 0;
921
```

Submitted

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